

BNO085 Embedded Heading Sensor System

STM32F767ZI + FreeRTOS + NMEA 0183 + ROS2 lifecycle driver

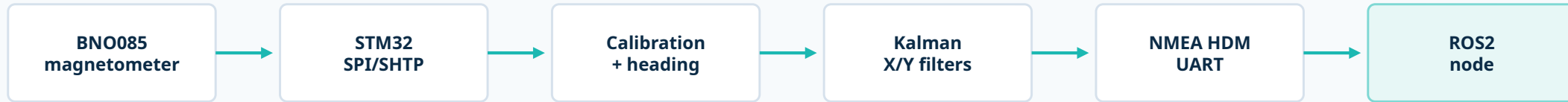
Objective

- Acquire calibrated magnetic-field data from BNO085
- Calculate magnetic heading and reduce fluctuation on STM32
- Transmit standard NMEA HDM sentences over UART
- Publish ROS2 IMU orientation, TF and diagnostics

Deliverable focus: reliable sensor acquisition, timing validation and ROS2 integration.

End-to-End System Architecture

From BNO085 calibrated magnetic field to ROS2 orientation and diagnostics.



NMEA output

```
furkan@Furkan: ~/Projects/Kuartis/ros2_ws
$HCHDM,146.98,M*1B
$HCHDM,147.26,M*1F
$HCHDM,147.48,M*17
$HCHDM,148.49,M*19
$HCHDM,149.02,M*17
$HCHDM,148.49,M*19
$HCHDM,148.32,M*15
$HCHDM,147.97,M*15
$HCHDM,147.93,M*10
```

\$HCHDM,<heading>,M*checksum

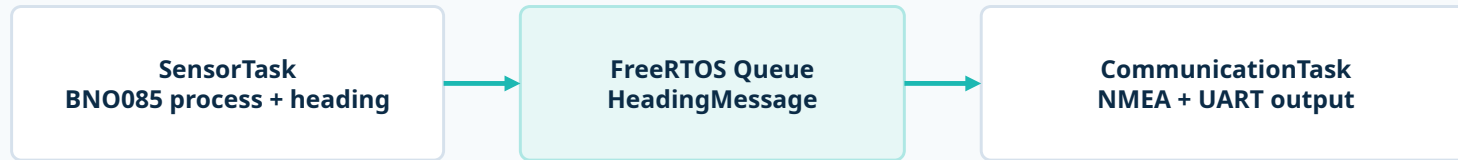
Key interfaces

- BNO085 communication through SPI and SHTP packets
- UART stream uses readable NMEA 0183 HDM format
- ROS2 converts heading into yaw quaternion and TF

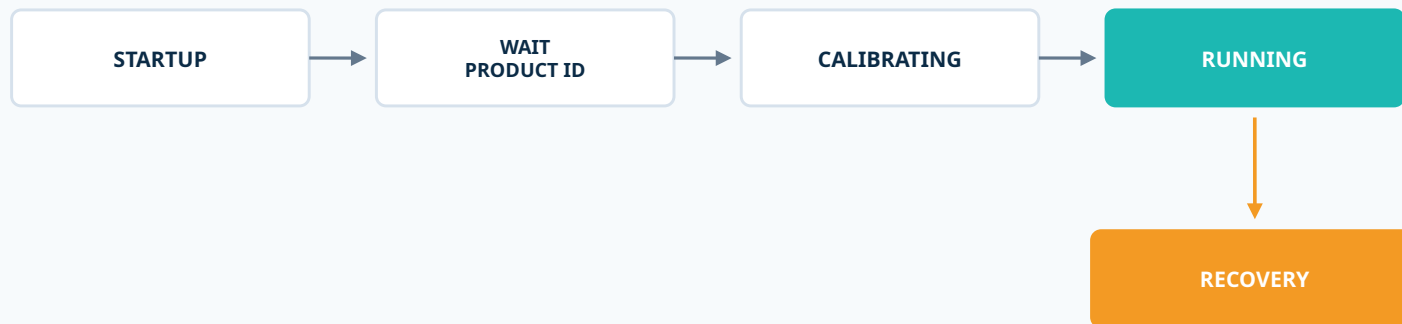
Embedded Runtime Architecture

FreeRTOS tasks isolate sensor acquisition from communication output.

Task-level design

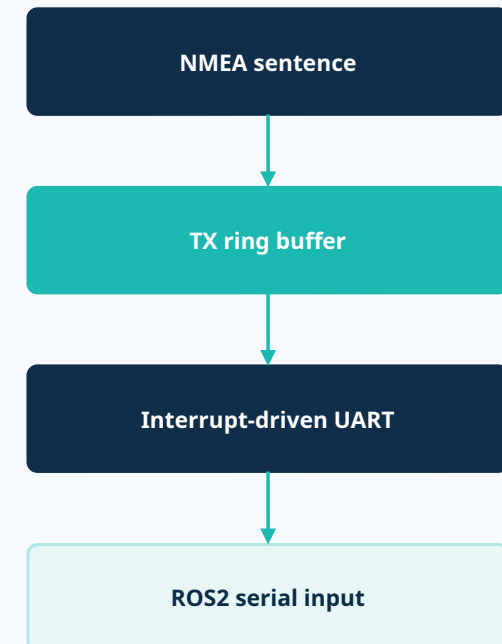


Sensor state machine



Timeouts and communication errors trigger recovery and re-initialization.

UART output path



Complete messages are enqueued to avoid partial NMEA frames.

Heading Calculation & Kalman Filtering

Heading is calculated from calibrated X/Y magnetic components and stabilized before ROS output.

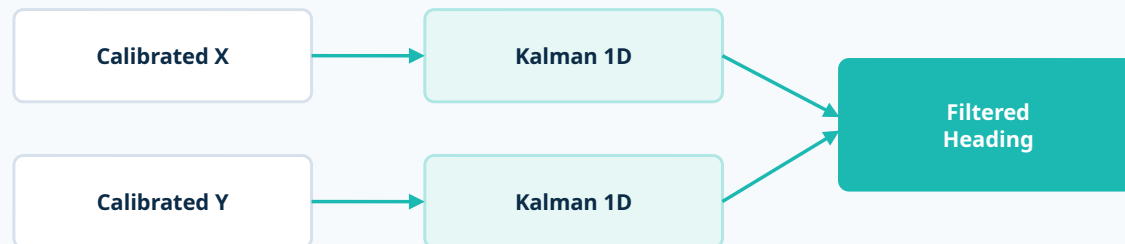
Heading calculation

$$\text{heading} = \text{atan2}(Y, X) \times 180 / \pi$$

if heading < 0° → heading = heading + 360°

The output is a magnetic heading in the 0° to 360° range.

Filtering strategy



Kalman parameter selection

Measurement noise R

Stationary magnetic-field variance was estimated with Welford algorithm.

$$R_x = 0.433 \mu T^2 \cdot R_y = 0.393 \mu T^2$$

Process noise Q = 0.25 $\mu T^2/s$ was tuned for smoothness vs responsiveness.

Core update equations

$$K = P^- / (P^- + R)$$

$$\mathbf{x} = \mathbf{x}^- + K(\mathbf{z} - \mathbf{x}^-)$$

Kalman gain is recalculated at every update.

Performance Validation

Embedded timing and ROS2 publish rate were measured separately.

Embedded timing counters

comm_timing_debug.sample_count	uint32_t	506
comm_timing_debug.current_period_ms	uint32_t	100
comm_timing_debug.min_period_ms	uint32_t	99
comm_timing_debug.max_period_ms	uint32_t	101
comm_timing_debug.average_period_ms	float	99.9743118
comm_timing_debug.frequency_hz	float	10.0025692
comm_timing_debug.peak_to_peak_jitter_ms	float	2
+ Add new expression		

Embedded result: 99-101 ms period, 2 ms peak-to-peak jitter, 10.003 Hz.

ROS2 publish rate

```
furkan@Furkan: ~/Projects/Kuartis... x furkan@Furkan: ~/Projects/Kuartis... x v
average rate: 10.003
min: 0.095s max: 0.105s std dev: 0.00052s window: 754
average rate: 10.003
min: 0.095s max: 0.105s std dev: 0.00052s window: 764
average rate: 10.003
min: 0.095s max: 0.105s std dev: 0.00051s window: 775
average rate: 10.003
min: 0.095s max: 0.105s std dev: 0.00051s window: 786
average rate: 10.002
min: 0.095s max: 0.105s std dev: 0.00051s window: 796
average rate: 10.002
min: 0.095s max: 0.105s std dev: 0.00050s window: 807
average rate: 10.002
min: 0.095s max: 0.105s std dev: 0.00050s window: 817
average rate: 10.002
min: 0.095s max: 0.105s std dev: 0.00050s window: 827
average rate: 10.002
min: 0.095s max: 0.105s std dev: 0.00050s window: 838
average rate: 10.002
min: 0.095s max: 0.105s std dev: 0.00049s window: 849
```

ROS2 result: /imu/data remains at approximately 10 Hz over 849 samples.

ROS2 Integration Outputs

Lifecycle node publishes standard ROS interfaces and runtime health information.

Diagnostics output

```
furkan@Furkan: ~/Projects/Kuartis/ros2_ws
furkan@Furkan: ~/Projects/Ku... x furkan@Furkan: ~/Projects/Ku... x furkan@
furkan@Furkan:~/Projects/Kuartis/ros2_ws$ ros2 topic echo /diagnostics
header:
  stamp:
    sec: 1788907219
    nanosec: 537141261
  frame_id: ''
status:
- level: '\0'
  name: 'bno085_driver: BNO085 Driver'
  message: BNO085 connected
  hardware_id: BNO085_STM32
  values:
    - key: Connection
      value: Connected
    - key: Frequency (Hz)
      value: '9.99958'
```

Connection: Connected, frequency: ~10 Hz.

/imu/data message

```
furkan@Furkan: ~/Projects/Kuartis/ros2_ws
furkan@Furkan: ~/Projects/Ku... x furkan@Furkan: ~/Projects/Ku... x furkan@Fur... x furkan@Fur... x
furkan@Furkan:~/Projects/Kuartis/ros2_ws$ ros2 topic echo /imu/data
header:
  stamp:
    sec: 1788907502
    nanosec: 834109035
  frame_id: imu_link
orientation:
  x: -0.0
  y: 0.0
  z: 0.9989386570101713
  w: -0.046060390040851946
orientation_covariance:
- 1000000.0
- 0.0
- 0.0
- 0.0
- 1000000.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.01
angular_velocity:
  x: 0.0
  y: 0.0
  z: 0.0
```

Heading is represented as a yaw quaternion.

TF output

```
At time 1788907724.679795388
- Translation: [0.000, 0.000, 0.000]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, 0.954, 0.299]
- Rotation: in RPY (radian) [0.000, -0.000, 2.534]
- Rotation: in RPY (degree) [0.000, -0.000, 145.160]
- Matrix:
-0.821 -0.571 0.000 0.000
0.571 -0.821 0.000 0.000
0.000 0.000 1.000 0.000
0.000 0.000 0.000 1.000
At time 1788907725.679813574
- Translation: [0.000, 0.000, 0.000]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, 0.865, 0.502]
- Rotation: in RPY (radian) [0.000, -0.000, 2.089]
- Rotation: in RPY (degree) [0.000, -0.000, 119.690]
- Matrix:
-0.495 -0.869 0.000 0.000
0.869 -0.495 0.000 0.000
0.000 0.000 1.000 0.000
0.000 0.000 0.000 1.000
```

base_link to imu_link transform exposes yaw.

Challenges, Improvements & Live Demonstration

The final step validates the complete embedded-to-ROS data path live.

Challenge

Magnetometer data still showed short-term heading fluctuation after calibration.

Cause: measurement noise and environmental magnetic disturbance.

Solution

Apply independent scalar Kalman filters to calibrated X and Y before heading calculation.

R was measured from stationary variance; Q was tuned experimentally.

Improvement

Add tilt compensation to improve heading reliability when the sensor is not level.

A future version can fuse accelerometer/orientation data for compensated heading.

Live demonstration sequence

1

CubeIDE debug

Check calibration_ok and runtime variables.

2

CubeMonitor graph

Compare raw heading and Kalman-filtered heading.

3

ROS2 topics

Show /imu/data content and publish rate.

4

TF + diagnostics

Show yaw transform and connected status.